FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Acute Upper Respiratory Tract Infections

Tutorial | Week 1 · Breathing, fever & the clinical methods | Paediatrics | Dr Aliku

Most upper respiratory tract infections in children are viral and need only fluids, paracetamol and a clear nose; the skill is to spot the few that need an antibiotic (streptococcal tonsillitis, to prevent rheumatic fever; acute otitis media; bacterial sinusitis), the ones that need surgery or an ENT referral (quinsy, mastoiditis, chronic ear discharge), and the airway emergencies (croup, epiglottitis, diphtheria) where a calm child, oxygen, steroids or adrenaline, and the most skilled airway person save lives.

Must know before the ward round

1. Common cold (coryza): viral, self-limiting 7–10 days; no antibiotics; saline drops and suction, fluids, paracetamol. No cough syrups under 6 years (UCG 2023); no honey under 1 year.
2. Sore throat: most are viral (cough, runny nose, hoarseness, conjunctivitis). Exudate + tender anterior neck nodes + fever + no cough in a child ≥3 y suggests group A streptococcus (Centor/McIsaac, FeverPAIN).
3. Strep tonsillitis: phenoxymethylpenicillin 10–20 mg/kg 6-hourly (max 500 mg) for 10 days, or benzathine penicillin IM single dose (600,000 IU if <30 kg; 1.2 MU if ≥30 kg; UCG: 30,000 IU/kg if <30 kg). A full 10 days (or the injection) prevents rheumatic fever.
4. Acute otitis media: ear pain, fever, red bulging drum or pus <14 days. Amoxicillin 40–45 mg/kg twice daily for 5 days (WHO/IMCI; AAP high-dose 80–90 mg/kg/day); UCG 2023 lists 15 mg/kg 8-hourly. Dry the ear by wicking.
5. Chronic suppurative otitis media: pus ≥14 days. Dry by wicking 3 times daily + quinolone (ciprofloxacin) ear drops for 2 weeks; systemic antibiotics are not useful; test for HIV.
6. Mastoiditis: tender swelling behind the ear pushing the pinna forward and down → admit, ceftriaxone 50–80 mg/kg IV daily + metronidazole 7.5 mg/kg 8-hourly, urgent ENT referral.
7. Croup: barking cough, hoarse voice, inspiratory stridor, 6 months–3 years, parainfluenza (and measles in Uganda). Dexamethasone 0.15–0.6 mg/kg orally once (UCG: 0.15 mg/kg; 0.6 mg/kg max 12 mg if severe) or prednisolone 1–2 mg/kg.
8. Severe croup: nebulised adrenaline 1:1000, 0.4–0.5 mL/kg (max 5 mL) undiluted or with saline, repeat after 30 min if needed; observe at least 2–4 h for rebound; oxygen; call the airway expert.
9. Epiglottitis: toxic, drooling, tripod, muffled voice, little cough. Do NOT examine the throat, lie the child down or give IM injections; secure the airway in theatre; ceftriaxone 50–100 mg/kg IV once daily for 7–10 days. Hib vaccine (pentavalent) has made it rare.
10. Diphtheria: grey adherent pseudomembrane that bleeds when removed, bull neck, low fever, in an unimmunised child. Give diphtheria antitoxin early (20,000–100,000 IU by severity) plus penicillin or a macrolide for 14 days; isolate; prophylaxis and vaccination for contacts; watch for myocarditis.
11. Never examine the throat with a spatula in a child with stridor and drooling; never sedate a child with upper airway obstruction.
12. Antibiotics do not prevent pneumonia after a cold; over-prescription drives resistance (WHO AWaRe: amoxicillin and penicillin V are Access antibiotics).

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In a nutshell

Upper respiratory tract infections (URTIs) are infections above the larynx (nose, sinuses, pharynx, tonsils, middle ear) plus the **laryngeal and tracheal** infections (croup, epiglottitis, bacterial tracheitis, diphtheria) that cause **stridor**. A Ugandan child has about **4–8 colds a year**, and URTIs are among the most common reasons for an outpatient visit, yet **almost all are viral**.

The clinician's job is to sort the child into one of four groups:

1. **Viral and self-limiting** (common cold, viral pharyngitis, most sore throats): reassure, symptom relief, safety-net advice, **no antibiotics**.
2. **Bacterial and treatable with a short oral course** (streptococcal tonsillitis, acute otitis media, bacterial sinusitis): amoxicillin or penicillin, given for the **right length of time**.
3. **Needs a procedure or specialist** (peritonsillar abscess, mastoiditis, chronic ear discharge, foreign body).
4. **Airway emergency** (severe croup, epiglottitis, diphtheria, bacterial tracheitis): keep the child calm, oxygen, steroid or adrenaline, and **the most skilled airway person**, now.

> **CLINICAL PEARL**
> The one thing never to forget: **a child with stridor and drooling must not be upset.** No spatula in the mouth, no lying flat, no IM injection, no separating from the mother. Agitation can turn a narrow airway into a closed one.

Definitions & classification

Syndrome	Site	Usual cause	Key feature
Common cold (coryza, rhinitis, rhinopharyngitis)	Nose and pharynx	Rhinovirus, RSV, adenovirus, coronavirus	Runny/blocked nose, sneezing, mild fever, 7–10 days
Pharyngitis / tonsillitis (pharyngotonsillitis)	Pharynx, tonsils	Viruses (most); **group A β-haemolytic streptococcus (GAS, *S. pyogenes*)** in 15–30% of school-age children	Sore throat, painful swallowing, ± exudate
Peritonsillar abscess (quinsy)	Between tonsil capsule and pharyngeal wall	GAS, anaerobes	Trismus, muffled "hot-potato" voice, uvula pushed to the opposite side
Acute otitis media (AOM)	Middle ear	*S. pneumoniae*, non-typeable *H. influenzae*, *M. catarrhalis*, viruses	Ear pain, fever, red **bulging** drum, or pus <14 days
Otitis media with effusion (glue ear)	Middle ear	After AOM, adenoids	Painless hearing loss, dull drum
Chronic suppurative otitis media (CSOM)	Middle ear with perforation	*Pseudomonas*, *S. aureus*, mixed	Ear discharge **≥14 days** (IMCI/UCG)
Mastoiditis	Mastoid air cells	Complication of AOM/CSOM	Tender swelling behind the ear, pinna pushed forward

Syndrome	Site	Usual cause	Key feature
Acute sinusitis	Paranasal sinuses	Viral; bacterial (pneumococcus, *H. influenzae*, *M. catarrhalis*)	Nasal discharge/cough **>10 days**, severe onset, or "double sickening"
Croup (laryngotracheobronchitis)	Larynx, trachea (subglottis)	**Parainfluenza**, RSV, influenza, **measles**	Barking cough, hoarseness, inspiratory stridor
Epiglottitis (supraglottitis)	Epiglottis, supraglottis	***H. influenzae* type b** (now rare after Hib vaccine), *S. pyogenes*, pneumococcus, *S. aureus*	Toxic, drooling, tripod posture, soft stridor, little cough
Bacterial tracheitis	Trachea	*S. aureus*, GAS	Croup that suddenly worsens, toxic, thick tracheal pus
Diphtheria	Tonsils, pharynx, larynx, nose, skin	Toxin-producing *Corynebacterium diphtheriae*	Grey adherent membrane, bull neck, myocarditis, neuropathy

IMCI classification for cough or cold (child 2 months–5 years)

Classification	Signs	Action
Severe pneumonia or very severe disease	Any general danger sign, or stridor in a calm child, or SpO_2 <90%	Pre-referral antibiotic, refer urgently
Pneumonia	Fast breathing and/or chest indrawing	Amoxicillin (see LRTI topic)
No pneumonia: cough or cold	None of the above	Home care, soothe the throat, **no antibiotics**; return if worse; follow up in 5 days if not improving; cough **>14 days** or recurrent wheeze → assess for TB, asthma, pertussis

The IMCI **ear problem** box: **mastoiditis** (tender swelling behind the ear) → refer; **acute ear infection** (pus <14 days or ear pain) → amoxicillin 5 days, wick the ear; **chronic ear infection** (pus ≥14 days) → wick, topical quinolone drops 2 weeks; **no ear infection** → no treatment.

Croup severity (clinical, with the Westley score)

Grade	Features	Westley score
Mild	Barking cough, **no stridor at rest**, little or no recession, playing and feeding	≤2
Moderate	**Stridor at rest**, visible recession, not agitated	3–7
Severe	Stridor at rest with **marked recession**, agitation or lethargy, tachycardia	8–11
Impending respiratory failure	Cyanosis, **decreased consciousness**, stridor becoming softer or biphasic, poor air entry, exhaustion	≥12

Westley components: level of consciousness (0 or 5), cyanosis (0, 4 or 5), stridor (0–2), air entry (0–2), recession (0–3).

Epidemiology & Uganda context

- Children under five have **4–8 viral URTIs a year**; more in day care, crowded homes and households with smoke.

- **Antibiotic overuse**: in Ugandan outpatient surveys most children with a cough or cold leave with an antibiotic (often amoxicillin or cotrimoxazole). Over-prescription drives resistance and side effects without benefit. **Amoxicillin and penicillin V are WHO AWaRe "Access" antibiotics**: when an antibiotic is needed, these are usually the right ones.
- **Rheumatic heart disease (RHD)**: Uganda has one of the highest burdens in the world; echocardiographic screening studies in Ugandan schoolchildren have found RHD in roughly **2–3 per 100**. Each case began with a group A streptococcal throat infection that was not treated (see the rheumatic fever topic). Treating strep throat with a full course of penicillin is **primary prevention**.
- **Hib** vaccine has been in Uganda's pentavalent vaccine (DTP-HepB-Hib) since **2002**, and **PCV** since **2013**, given at **6, 10 and 14 weeks**: epiglottitis and pneumococcal ear disease have fallen sharply. An unvaccinated child with epiglottitis is a missed opportunity.
- **Diphtheria** still occurs in under-immunised communities (and in refugee settlements and border districts); DTP3 coverage in Uganda is high nationally but uneven by district.
- **Measles** causes croup, otitis media and pneumonia; Uganda gives **measles-rubella at 9 and 18 months** (UNEPI) and has ongoing outbreaks (MoH 2026).
- **CSOM** ("milk in the ear", as some families call it) is common, especially in HIV-positive and malnourished children; it is a leading cause of preventable hearing loss.

> **UGANDA CONTEXT**
> Current UNEPI schedule highlights relevant to URTIs: **BCG, OPV0 and hepatitis B birth dose** (introduced 2025) at birth; **DTP-HepB-Hib, PCV, OPV** at **6, 10, 14 weeks** (plus rotavirus and IPV); **measles-rubella at 9 and 18 months**; yellow fever at 9 months; **malaria vaccine at 6, 7, 8 and 18 months** in high-burden districts (since April 2025). Always check the child's card.

Pathophysiology

Why children get so many URTIs and ear infections

- Immature immunity, crowding and smoke exposure.
- **The Eustachian tube** in infants is **shorter, wider and more horizontal**, so nasopharyngeal secretions and bacteria reach the middle ear easily after a cold; large adenoids block it. That is why **AOM follows a cold** in a toddler.
- Bottle feeding lying flat and exposure to smoke increase AOM; breastfeeding protects.

Viral vs bacterial throat infections

- Viruses inflame the whole upper airway: **cough, runny nose, hoarseness, conjunctivitis, ulcers**.
- **GAS** causes a localised exudative tonsillitis with **tender anterior cervical nodes, fever, and no cough**, in children **3–15 years** (rare under 3).
- **Molecular mimicry**: antibodies against streptococcal M protein cross-react with heart valves, joints and brain → **acute rheumatic fever** 2–4 weeks later. Treating within **9 days** of the sore throat prevents most ARF. Post-streptococcal **glomerulonephritis** is not prevented by antibiotics.

Why croup and epiglottitis obstruct a child's airway

- The **subglottis** (cricoid ring) is the narrowest part of a child's airway and is a complete ring of cartilage: mucosal swelling can only narrow the lumen.
- Resistance $\propto 1/\text{radius}^4$ (Poiseuille): **1 mm of oedema** in a 4 mm infant subglottis increases resistance about **16-fold**; in an adult, much less.

- Crying increases turbulent flow and negative pressure, which collapses the floppy extrathoracic airway further: **keep the child calm**.
- **Steroids** reduce mucosal oedema (onset 30 min–2 h, last 24–48 h). **Adrenaline** causes α-mediated vasoconstriction of the mucosa (onset 10 min, lasts about 2 h; symptoms can **rebound**).
- **Epiglottitis** is cellulitis of the supraglottis: the swollen epiglottis can fall back and **completely obstruct** the airway if the child lies down or gags.

Diphtheria

C. diphtheriae infected by a bacteriophage produces an **exotoxin** that kills epithelial cells (forming the **pseudomembrane** of dead cells, fibrin and bacteria) and enters the blood to damage the **heart (myocarditis, days 7–14)** and **nerves (palatal palsy, then cranial and peripheral neuropathy, weeks 2–8)**. **Antitoxin neutralises only unbound toxin**, so it must be given **early**, on clinical suspicion.

Clinical acumen

Focused history

- **Main symptom and duration**: runny nose, sore throat, ear pain or discharge (how long? ≥14 days = chronic), noisy breathing, barking cough, drooling, voice change.
- **Fever pattern**: high fever and toxicity (epiglottitis, tracheitis, quinsy, mastoiditis); low-grade (croup, diphtheria).
- **Onset**: hours (epiglottitis, foreign body, anaphylaxis); 1–2 days after a cold (croup); sudden choking (foreign body).
- **Eating and drinking**: refusal to swallow saliva = drooling (epiglottitis, quinsy, retropharyngeal abscess).
- **Immunisation card**: DTP-HepB-Hib ×3, PCV ×3, MR ×2 (UNEPI). Unvaccinated → think diphtheria, epiglottitis, measles.
- **Contacts**: sick siblings, school outbreaks, measles in the village, diphtheria contacts.
- **Previous sore throats, joint pains, heart murmur** (ARF/RHD), previous ear discharge.
- **HIV status** (recurrent AOM, CSOM, oral candida), **nutrition** and growth (Z-scores), smoke exposure, day care.
- **Drugs given**: antibiotics already taken (how long?), herbal treatments, "uvulectomy" or tonsil cutting by traditional healers (bleeding, infection).

Examination

1. **From the doorway**: posture (tripod/sniffing), drooling, stridor at rest, recession, colour, alertness, voice (hoarse, muffled), toxicity.
2. **Vital signs**: temperature, RR, HR, **SpO_2**.
3. **If no airway concern**: look in the throat with a torch (tonsil size, exudate, membrane, uvula deviation, petechiae on the palate), **neck nodes** (tender anterior chain in GAS; bull neck in diphtheria), **ears with an otoscope** (drum colour, bulging, perforation, discharge), sinuses (tenderness, periorbital swelling), nose (discharge, foreign body: unilateral foul discharge).
4. **Chest**: exclude pneumonia (fast breathing, indrawing, crackles).
5. **Skin**: scarlet fever rash (sandpaper), measles rash, Koplik spots; **heart** for murmurs (RHD, diphtheritic myocarditis: arrhythmia, gallop).

Red flags

RED FLAG

Airway danger: stridor at rest, drooling, tripod posture, toxic look, cyanosis or SpO_2 <92%, decreasing consciousness, a soft or disappearing stridor in a tiring child, a grey membrane in the throat, a bull neck. Call the most senior clinician and anaesthetist/ENT, keep the child with the mother, give oxygen without upsetting the child.

RED FLAG

Spread beyond the ear or sinus: swelling behind the ear, headache, neck stiffness, drowsiness or convulsions (meningitis, brain abscess), facial palsy, eye swelling with painful or reduced eye movement or proptosis (orbital cellulitis). Admit, IV antibiotics, refer.

Thinking like a clinician

- **"Cough + runny nose + sore throat = viral."** The more sites involved, the more likely viral.
- **"Barking cough = croup; drooling without cough = epiglottitis."** Croup children cough and want to drink; epiglottitis children sit still, drool and avoid swallowing.
- **"Croup that suddenly gets worse with high fever and toxicity"** = **bacterial tracheitis** (does not respond to adrenaline): needs airway and IV antibiotics.
- **Stridor in an unimmunised child with a grey membrane** = **diphtheria** until proven otherwise: give antitoxin on clinical suspicion; don't wait for culture.
- **Under 3 years with exudative tonsillitis** is usually viral (adenovirus, EBV); GAS is uncommon.
- **Exudative tonsillitis + generalised lymphadenopathy + splenomegaly** in a teenager = **infectious mononucleosis**; amoxicillin causes a rash.
- **Unilateral foul-smelling nasal discharge** in a toddler = **foreign body** in the nose.
- **Ear discharge ≥14 days** = CSOM: oral antibiotics don't help; drying and topical quinolone drops do. Test for HIV.
- **"No antibiotics" is a treatment**, and it needs an explanation and a safety net: what to watch for and when to come back.

Clinical workup

Step 0: First 5 minutes

1. **ETAT triage**: is there **obstructed breathing (stridor at rest), severe respiratory distress, central cyanosis**? These are **emergency signs**. Move to the resuscitation area.
2. **Do not upset the child**: examine on the mother's lap; leave the throat alone if there is stridor or drooling.
3. **Oxygen** (blow-by or mask held by the mother) if SpO_2 <92% or severe distress.
4. **Decide the airway pathway**:
    - Barking cough + stridor → **croup** → steroid now; **nebulised adrenaline** if severe.
    - Toxic + drooling + tripod → **epiglottitis** → call anaesthetist/ENT, theatre, no interventions until the airway is secure.
    - Grey membrane, unimmunised → **diphtheria** → isolate, antitoxin, antibiotics, airway team.
    - Sudden choking → **foreign body**: if the child can cough, encourage coughing; if not effective, back blows/chest thrusts (infant) or abdominal thrusts (child >1 y).
    - Urticaria, angioedema, hypotension → **anaphylaxis** → IM adrenaline 0.01 mg/kg.

5. **If no airway emergency**: check for **general danger signs** (unable to drink/breastfeed, vomits everything, convulsions, lethargic/unconscious) and fast breathing or chest indrawing (pneumonia: see the LRTI topic).

Step 1: Focused history

- [ ] **Duration** → cold <10 days = viral; nasal symptoms >10 days = possible bacterial sinusitis; ear pus ≥14 days = CSOM; cough >14 days = TB, asthma, pertussis.
- [ ] **Cough present?** → yes favours viral (and croup); absent favours GAS and epiglottitis.
- [ ] **Drooling, voice change, trismus** → epiglottitis, quinsy, retropharyngeal abscess.
- [ ] **Immunisation card** → missed DTP/Hib/PCV/MR changes the differential (diphtheria, epiglottitis, measles).
- [ ] **Previous rheumatic fever/RHD** → any sore throat must be treated; check secondary prophylaxis.
- [ ] **HIV status, growth, feeding** → recurrent and chronic ear disease; need for cotrimoxazole/ART.
- [ ] **Antibiotics already taken** → failure of amoxicillin after 48–72 h in AOM → amoxicillin-clavulanate.
- [ ] **Contacts** → diphtheria, measles, epiglottitis (Hib household prophylaxis).

Step 2: Focused examination

General: toxicity, hydration, posture, drooling, voice, stridor (at rest or only when upset), recession, SpO_2, temperature.

ENT (only if no airway threat): nose (discharge, foreign body), throat (tonsil size, exudate, membrane, uvula), neck (nodes, bull neck, torticollis in retropharyngeal abscess), ears (otoscopy), sinuses, eyes (periorbital swelling).

Chest and heart: fast breathing, crackles, murmurs, arrhythmia.

Document these positives	And these negatives
Stridor at rest or not; recession; SpO_2	"No drooling, no toxicity; drinking well"
Tonsils: size, exudate; nodes: tender anterior	"No membrane; uvula central; no trismus"
Ear drum: colour, bulging, perforation, discharge duration	"No mastoid swelling or tenderness; no facial palsy"
Immunisation status	"No fast breathing or chest indrawing"

Step 3: Bedside tests

Test	How	What it tells you
Pulse oximetry	Correct probe size, calm child	<92%: severe croup or airway compromise; oxygen and senior review
Croup score (Westley)	Consciousness, cyanosis, stridor, air entry, recession	Grades severity; repeat after steroid/adrenaline
Centor/McIsaac or FeverPAIN score	See Management	Probability of GAS; whether to treat
Rapid antigen detection test (RADT) for GAS	Throat swab, 10 minutes	Positive → treat; not widely available in Uganda (private hospitals)
Otoscopy	See the respiratory/ENT exam topic	Bulging drum (AOM) vs dull retracted drum (effusion) vs perforation (CSOM)
Malaria RDT	Every febrile child in Uganda	Fever may be malaria, not the throat
RBS	Sick or not feeding	Hypoglycaemia

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Throat swab culture	Suspected GAS where available; outbreaks; **suspected diphtheria** (tell the lab: needs tellurite medium)	GAS; *C. diphtheriae* (toxin testing at the national reference lab, UNHLS/CPHL)	Confirms; but **treat clinically** first. Culture at RRH/NRH only
Ear swab culture	CSOM not responding, mastoiditis	*Pseudomonas*, *S. aureus*, Proteus	Guides topical/systemic choice. RRH/NRH
Full blood count	Toxic child, mastoiditis, quinsy	Neutrophilia; atypical lymphocytes (mononucleosis)	Supports bacterial cause; not needed for simple URTI
ASOT	Suspected ARF or post-streptococcal GN (not for sore throat)	Raised	Evidence of recent GAS infection
Lateral neck X-ray	**Only** if stable and diagnosis unclear, never in suspected epiglottitis with distress	"Thumb sign" (epiglottitis), "steeple sign" on AP (croup), widened prevertebral space (retropharyngeal abscess), radiopaque foreign body	Diagnostic clue; never delay airway care for a film
ECG	**Diphtheria** (daily in the first 2 weeks)	ST-T changes, heart block, arrhythmia	Myocarditis: bed rest, cardiac monitoring
CT head/mastoid/sinus	Mastoiditis or sinusitis with intracranial or orbital complications	Abscess, sinus thrombosis	Surgery; neurosurgery. NRH and private hospitals
HIV test	CSOM, recurrent AOM, oral thrush, poor growth	Positive	ART, cotrimoxazole
Audiometry/tympanometry	Glue ear, CSOM, after mastoiditis	Conductive hearing loss	ENT referral, school support. Mulago/Mengo ENT clinics

Step 5: Putting it together

Model one-line summary

> "A 2-year-old boy weighing 11 kg (WAZ −0.8), fully immunised, HIV-exposed but negative at 18 months, with 2 days of coryza followed by a barking cough, hoarse voice and stridor at rest since midnight, alert, SpO_2 95%, moderate subcostal recession, no drooling and drinking well: **moderate croup**."

Problem list

1. Moderate viral croup (Westley 4).
2. Mild fever, drinking well.
3. Immunisations complete for age (check MR2 due at 18 months: given).

Working diagnosis: *acute laryngotracheobronchitis (croup), moderate severity.*

EXAM FAVOURITE

Worked example 1: scoring a sore throat. A 9-year-old girl: fever 38.6 °C, exudate on both tonsils, tender anterior cervical nodes, **no cough**. McIsaac score: fever 1 + exudate 1 + nodes 1 + no cough 1 + age 3–14 y 1 = **5** → high probability of GAS (about 50%). In Uganda, with a high RHD burden and no rapid test, **treat**: benzathine penicillin 600,000 IU IM once (weight 26 kg) or penicillin V for 10 days. A 4-year-old with runny nose, cough, hoarse voice and red throat scores 0–1 → viral, no antibiotic.

EXAM FAVOURITE

Worked example 2: the croup score after treatment. A 14-month-old (9 kg): stridor at rest (2), moderate recession (2), decreased air entry (1), agitated but no cyanosis (0) → **Westley 5, moderate**. Given dexamethasone 0.6 mg/kg (5.4 mg) orally. At 30 minutes: stridor at rest louder, marked recession (3), SpO_2 91% → score now 6–7 and worsening → **nebulised adrenaline 1:1000, 4.5 mL**, oxygen. At 20 minutes: no stridor at rest. **Must stay at least 2–4 hours** because adrenaline wears off; admit if stridor returns.

Differential diagnosis

Of stridor (acute)

Condition	Distinguishing features	Key investigation
Croup	6 m–3 y, coryza 1–2 days, **barking cough**, hoarse, mild fever, not toxic, wants to drink	Clinical
Epiglottitis	2–7 y (any age if unvaccinated), **sudden high fever, toxic, drooling, tripod**, muffled voice, **little or no cough**	Clinical; airway in theatre
Bacterial tracheitis	Starts like croup then **sudden deterioration**, high fever, toxic, copious secretions, no response to adrenaline	Bronchoscopy (pus, membrane)
Diphtheria	Unimmunised, low fever, **grey adherent membrane**, bull neck, hoarseness, myocarditis later	Throat swab culture; ECG
Inhaled foreign body (laryngeal)	Sudden choking, no fever, any age (toddlers)	X-ray (if radiopaque), laryngoscopy
Retropharyngeal abscess	<5 y, fever, neck stiffness or torticollis, drooling, muffled voice	Lateral neck X-ray (widened prevertebral space), CT
Anaphylaxis / angioedema	Trigger, urticaria, lip/tongue swelling, wheeze, hypotension	Clinical
Measles croup	Rash, conjunctivitis, Koplik spots, unvaccinated	Clinical
Laryngomalacia (chronic)	Stridor from early weeks, worse lying supine/feeding, thriving	Laryngoscopy (ENT)

Of sore throat

Condition	Distinguishing features	Key investigation
Viral pharyngitis	Cough, coryza, conjunctivitis, hoarseness, ulcers	Clinical
GAS tonsillitis	3–15 y, fever, exudate, tender anterior nodes, no cough, palatal petechiae, scarlet rash	RADT/culture where available
Infectious mononucleosis (EBV)	Teenager, exudate, posterior nodes, **splenomegaly**, fatigue	FBC (atypical lymphocytes), monospot

Condition	Distinguishing features	Key investigation
Diphtheria	Grey membrane that bleeds when removed, bull neck, unimmunised	Culture
Peritonsillar abscess	Trismus, uvula deviated, hot-potato voice	Clinical; aspiration
Oral candidiasis / herpes gingivostomatitis	White plaques (HIV, antibiotics) / anterior mouth ulcers and gingivitis	Clinical; HIV test

Of ear pain or discharge

Condition	Distinguishing features	Key investigation
Acute otitis media	Pain, fever, red bulging drum; pus <14 days after perforation	Otoscopy
Otitis externa	Pain on moving the pinna, swollen canal, drum normal	Otoscopy
CSOM	Painless discharge ≥14 days, perforation	Otoscopy, swab
Mastoiditis	Post-auricular swelling, pinna pushed forward and down	Clinical; CT
Foreign body	Toddler, unilateral, visible object	Otoscopy
Referred pain	Tonsillitis, dental abscess, teething	Examine throat and teeth

Investigations

Test	What you are looking for	Why
Clinical examination (throat, ears, neck)	Exudate, bulging drum, membrane, mastoid swelling	Most URTIs need no laboratory test
SpO_2	<92% in croup/epiglottitis	Severity, need for oxygen
Throat swab (culture/RADT)	GAS; *C. diphtheriae*	Targets antibiotics; public health action
Ear swab	Organisms in CSOM	Resistant infection
Lateral neck X-ray (stable child only)	Thumb sign, prevertebral swelling	Epiglottitis vs retropharyngeal abscess
ECG (diphtheria)	Myocarditis	Predicts death; guides monitoring
HIV test	Infection	Recurrent/chronic ear disease

Management

A. Principles and antibiotic stewardship

1. **Most URTIs need no antibiotic.** Explain why (viral; antibiotics don't shorten it or prevent pneumonia; they cause diarrhoea, rash and resistance).
2. **Symptom relief**: paracetamol **15 mg/kg** 6-hourly (max 60 mg/kg/day) or ibuprofen 5–10 mg/kg 8-hourly (with food; avoid if dehydrated); fluids; saline nose drops; continue breastfeeding.
3. **Safety net**: return immediately if fast or difficult breathing, stridor at rest, unable to drink, very sleepy, convulsions, ear swelling, or not better in 5 days.
4. **When an antibiotic is needed, use the narrowest effective one** (penicillin V, amoxicillin) for the recommended time.

B. Common cold (UCG 2023 5.2.3)

Do	Don't
Continue breastfeeding; extra fluids; warm drinks	Antibiotics
Clear the nose with **saline drops** and gentle suction (bulb or cloth wick) before feeds	**Cough syrups, antihistamines, decongestant mixtures in <6 y** (UCG 2023: avoid; risk of toxicity)
Paracetamol for 2–3 days for fever or pain	Honey in **<1 y** (infant botulism)
Steam/honey (>1 y) as home remedies	Xylometazoline drops in infants; if used in older children (0.05%) **max 5 days** (rebound congestion)
Advise when to return (danger signs, fast breathing, ear pain)	

C. Pharyngitis and tonsillitis (UCG 2023 21.3.3–21.3.4)

Who to treat with an antibiotic

Score	Items	Decision
McIsaac (modified Centor)	Fever >38 °C (+1); tonsillar exudate or swelling (+1); tender/swollen anterior cervical nodes (+1); **absence of cough** (+1); age 3–14 y (+1); age ≥45 (–1)	0–1: no antibiotic; 2–3: test if possible, otherwise clinical judgement; **4–5: treat**
FeverPAIN (NICE NG84)	**F**ever in past 24 h; **P**urulence; **A**ttend rapidly (≤3 days); **I**nflamed tonsils; **N**o cough or coryza	0–1 no antibiotic; 2–3 consider delayed; **4–5 antibiotic**

UGANDA CONTEXT

In Uganda, with a very high burden of rheumatic heart disease and few rapid strep tests, **treat clinically suspected streptococcal tonsillitis** in children aged 3–15 years (exudate, tender anterior nodes, fever, no cough), and **always treat** a child who has had rheumatic fever or has RHD. UCG 2023 recommends a **full 10-day course** (or a single benzathine injection). NICE's 5-day option is for symptom relief in a low-RHD country and is **not** appropriate here.

Antibiotic choices

Drug	Dose	Route	Frequency	Duration	Notes
Phenoxymethylpenicillin (penicillin V)	**10–20 mg/kg** per dose (UCG 2023), max 500 mg; or by age: 1–5 y 125 mg, 6–11 y 250 mg (NICE NG84 qds)	Oral	**6-hourly** (or 12-hourly at double dose)	**10 days**	First line; adherence falls after symptoms settle: stress the full course
Benzathine benzylpenicillin	**600,000 IU if <30 kg; 1,200,000 IU if ≥30 kg** (WHO/AHA; UCG 2023: 30,000 IU/kg if <30 kg, 1.2 MU adults)	**Deep IM**	Once	Single dose	Best when adherence is doubtful; painful (can be mixed with lidocaine 1% per local protocol). Have adrenaline ready
Amoxicillin	**50 mg/kg once daily** (max 1 g) or 25 mg/kg twice daily	Oral	Once/twice daily	**10 days**	Equally effective, tastes better; avoid if mononucleosis possible (rash)
Erythromycin (penicillin allergy)	**12.5 mg/kg** (max 500 mg)	Oral	6-hourly	10 days	UCG 2023 allergy alternative

Drug	Dose	Route	Frequency	Duration	Notes
Azithromycin (penicillin allergy)	**12 mg/kg** once daily (max 500 mg)	Oral	Once daily	**5 days**	GAS macrolide resistance exists

Peritonsillar abscess (quinsy): admit; IV benzylpenicillin 50,000 IU/kg 6-hourly (or ceftriaxone) **plus metronidazole 7.5 mg/kg 8-hourly**; analgesia; IV fluids if unable to swallow; **needle aspiration or incision and drainage by ENT**; then oral antibiotics to 10–14 days. Tonsillectomy after recurrent quinsy.

Recurrent tonsillitis: tonsillectomy is considered for ≥7 episodes in 1 year, ≥5/year for 2 years or ≥3/year for 3 years (Paradise criteria), sleep-disordered breathing, or recurrent quinsy (ENT referral).

D. Ear infections

Acute otitis media

1. **Analgesia for everyone**: paracetamol or ibuprofen for 2–3 days.
2. **Antibiotic** (WHO pocket book/IMCI: all children with AOM in our setting): **amoxicillin**.
3. **Dry the ear by wicking** if there is pus: twist a clean absorbent cloth or tissue into a wick, insert, remove when wet, replace; repeat until dry; 3 times a day. No cotton buds, no oils, no herbal drops.
4. **Review after 5 days**: pain and fever settled? Pus persists → continue wicking; recurrent → check HIV, hearing.
5. **Failure after 48–72 h**: amoxicillin-clavulanate (amoxicillin component 45 mg/kg twice daily) or ceftriaxone 50 mg/kg IM daily for 3 days.

> UGANDA CONTEXT
> Amoxicillin doses for AOM differ: **WHO pocket book/IMCI: 40 mg/kg twice daily for 5 days** (the same dispersible tablets as pneumonia); **UCG 2023: 15 mg/kg 8-hourly for 5 days** (≈45 mg/kg/day); **AAP 2013: high-dose 80–90 mg/kg/day** in 2 doses (10 days if <2 y) because of pneumococcal resistance. In Ugandan wards and OPDs, the **IMCI 40 mg/kg twice-daily dose** is what most clinicians use. Quote UCG in the exam and know that higher doses exist.

Chronic suppurative otitis media (pus ≥14 days)

- **Dry mopping (wicking) 3 times daily** until the ear is dry (may take weeks).
- **Topical quinolone drops** (ciprofloxacin ear drops) 2–4 drops after each wicking, for **2 weeks** (IMCI/WHO; UCG 2023).
- UCG 2023 also describes gentle aural irrigation with dilute hydrogen peroxide (1 spoon in half a glass of clean lukewarm water) before wicking.
- **Systemic antibiotics are not recommended** (UCG 2023) unless complications.
- **Keep water out** of the ear when bathing or swimming. **HIV test.** Hearing assessment and ENT referral if persistent, cholesteatoma (white debris, foul smell), or polyps.

Mastoiditis (UCG 2023 21.1.6): admit urgently; **ceftriaxone 50–80 mg/kg IV/IM once daily for 10–14 days** (max 2–4 g) **plus metronidazole 7.5 mg/kg 8-hourly**; analgesia; urgent ENT for incision and drainage/mastoidectomy; watch for meningitis, brain abscess, facial palsy, sigmoid sinus thrombosis.

Glue ear (otitis media with effusion): most resolves in 3 months; hearing check; ENT referral if persistent >3 months with hearing loss or speech delay (grommets ± adenoidectomy). UCG 2023 lists chlorphenamine and decongestant drops; international guidelines do not recommend antihistamines or decongestants for glue ear.

E. Acute sinusitis (UCG 2023 21.2.4)

- **Diagnose bacterial sinusitis only if**: nasal discharge or daytime cough **>10 days** without improvement; **severe onset** (fever ≥39 °C with purulent discharge for ≥3 days); or **worsening after initial improvement** ("double sickening") (AAP 2013; UCG: >1 week, unilateral facial pain, worsening).
- **Treatment**: saline irrigation, analgesia; **amoxicillin** 15 mg/kg 8-hourly (UCG) or 40–45 mg/kg twice daily (high-dose), for **7–10 days**; amoxicillin-clavulanate if no improvement in 72 h. Look for a **dental focus** or **nasal foreign body**.
- **Orbital complications** (periorbital swelling, pain on eye movement, proptosis, reduced vision): admit; **IV ceftriaxone 50–80 mg/kg daily + cloxacillin 50 mg/kg 6-hourly** (or + metronidazole); urgent ophthalmology/ENT and CT.

F. Croup (UCG 2023 5.2.7)

Steps

1. **Keep the child calm on the parent's lap**; no throat examination; no unnecessary procedures.
2. **Steroid for every child with croup who has stridor** (and for barking cough in young infants or those far from care):
   - **Dexamethasone 0.15 mg/kg orally once** (UCG 2023 mild croup; RCH mild–moderate), or **0.6 mg/kg (max 12 mg) orally/IM/IV** for severe croup (RCH; Medscape: trials used 0.6 mg/kg; 0.15 mg/kg is effective).
   - Or **prednisolone 1–2 mg/kg orally once** (UCG 2023; may need repeating).
3. **Severe croup** (stridor at rest with marked recession, agitation, SpO_2 <92%):
   - **Nebulised adrenaline 1:1000: 0.4–0.5 mL/kg (max 5 mL)**, made up to 4 mL with saline if needed; repeat after 30 minutes if necessary (UCG 2023: 0.4 mg/kg, max 5 mg).
   - **Oxygen** (humidified, held near the face by the parent) if SpO_2 <92%.
   - IV/IM steroid if unable to take oral: dexamethasone 0.3–0.6 mg/kg IM (UCG: 300 µg/kg IM), or hydrocortisone (UCG 2023: <1 y 25 mg, 1–5 y 50 mg, 6–12 y 100 mg) slow IV/IM; repeat after 6 h if necessary.
   - **Observe for at least 2–4 hours after adrenaline** (rebound).
4. **Impending respiratory failure** (cyanosis, decreasing consciousness, exhaustion): 100% oxygen, repeated adrenaline, call the anaesthetist; **intubation in theatre with a tube 0.5–1 mm smaller than usual**; tracheostomy if intubation fails (UCG: nasotracheal intubation or tracheostomy).
5. **Admit** if: stridor at rest after treatment, needed adrenaline (observe ≥4 h, admit if stridor returns), age <6 months, SpO_2 <92%, poor drinking, lives far away or unreliable return.
6. **Discharge**: no stridor at rest, drinking, parents understand when to return (stridor at rest, recession, drooling, blue colour, drowsiness).
7. **No antibiotics** unless bacterial tracheitis suspected (then treat as epiglottitis). **No cough mixtures** (UCG). Humidified air ("steam") has **no proven benefit** and risks scalds: do not advise steam kettles.

> **UGANDA CONTEXT**
> UCG 2023 advises IV hydration with **half-strength Darrow's in 2.5% glucose** for severe croup. Most Kampala wards now use **standard maintenance fluids** (e.g. Ringer's lactate or 0.9% saline with 5% glucose for younger children) only if the child cannot drink. Many children with croup in Uganda have **measles**: check for rash, Koplik spots and vaccination status, give vitamin A and isolate.

G. Epiglottitis (UCG 2023 5.2.4)

1. **Recognise it**: toxic, high fever, drooling, tripod/sniffing posture, muffled voice, soft stridor, anxious, **little or no cough**.
2. **Do NOT**: look in the throat, lie the child flat, take blood, give IM injections, or send the child to X-ray unaccompanied. Keep with the mother; oxygen held near the face.
3. **Call** the most experienced anaesthetist and ENT surgeon; **secure the airway in theatre** (gas induction, child sitting; tracheostomy set open). Bag-valve-mask ventilation usually works if the child obstructs before help arrives.
4. **After the airway is secure**: IV line, blood culture, **ceftriaxone 50–100 mg/kg IV once daily for 7–10 days** (UCG 2023: 50 mg/kg; many centres use 80–100 mg/kg, max 2 g) ± cloxacillin if *S. aureus* suspected.
5. **Hib prophylaxis**: rifampicin 20 mg/kg once daily (max 600 mg) for 4 days to household contacts if there is an unvaccinated or incompletely vaccinated child <4 years, or an immunocompromised child, in the home (CDC/AAP).
6. **Check and complete immunisation** (pentavalent).

H. Diphtheria (UCG 2023 2.1.3; WHO)

1. **Isolate** (droplet and contact precautions) until 2–3 negative cultures (UCG: 3 negative swabs) or 48 h of appropriate antibiotics.
2. **Diphtheria antitoxin (DAT)** as soon as diphtheria is suspected (don't wait for culture), after a test dose per the product leaflet (equine serum: anaphylaxis possible; adrenaline ready). Dose by severity (WHO): **20,000–40,000 IU** for pharyngeal/laryngeal disease <48 h; **40,000–60,000 IU** for nasopharyngeal disease; **80,000–100,000 IU** for extensive disease >3 days or bull neck; given IV in 0.9% saline over 1 h (or IM). WHO Pocket Book 2013: **40,000 units IM/IV**. DAT is scarce: request through MoH/national stores.
3. **Antibiotic** for 14 days:
   - UCG 2023: **procaine benzylpenicillin 50,000 IU/kg IM once daily** until able to swallow, then **penicillin V** (<1 y 12.5 mg/kg; 1–6 y 125 mg; >6 y 250 mg) 6-hourly to complete **14 days**.
   - Penicillin allergy: **erythromycin 12.5 mg/kg 6-hourly** (50 mg/kg/day) for 14 days. WHO's 2024 clinical guidance suggests a **macrolide (azithromycin or erythromycin)** as first choice where available.
4. **Airway**: watch for laryngeal extension and membrane aspiration; tracheostomy may be needed. Avoid pulling off the membrane.
5. **Heart**: strict bed rest, **ECG** daily for 2 weeks, monitor for arrhythmia and heart failure (myocarditis typically days 7–14).
6. **Nerves**: watch for palatal palsy (nasal regurgitation), then cranial, respiratory and limb weakness (weeks 2–8): NG feeding, ventilatory support.
7. **Contacts**: throat swabs; **single dose benzathine penicillin IM** (UCG: 600,000 IU if <10 y, 1.2 MU if ≥10 y) or erythromycin for 7–10 days; check and complete DTP vaccination, booster if last dose >1 year ago; monitor for 7 days.
8. **Notify** the district surveillance focal person immediately (epidemic-prone disease). **Vaccinate the patient** after recovery (disease does not reliably confer immunity).

I. Drug table summary

Drug	Indication	Dose	Route	Frequency	Duration	Max / notes
Paracetamol	Pain, fever	15 mg/kg	Oral	6-hourly	2–3 days	60 mg/kg/day

Drug	Indication	Dose	Route	Frequency	Duration	Max / notes
Ibuprofen	Pain	5–10 mg/kg	Oral	8-hourly with food	2–3 days	Avoid if dehydrated, asthma with NSAID sensitivity
Penicillin V	GAS tonsillitis	10–20 mg/kg	Oral	6-hourly	10 days	500 mg/dose
Benzathine penicillin	GAS tonsillitis; diphtheria contacts	600,000 IU (<30 kg); 1.2 MU (≥30 kg)	Deep IM	Once	Single	Anaphylaxis kit ready
Amoxicillin	AOM, sinusitis, GAS	40–45 mg/kg (WHO/IMCI; UCG 15 mg/kg 8-hourly)	Oral (dispersible)	12-hourly	5 days AOM; 7–10 days sinusitis; 10 days GAS (50 mg/kg once daily)	1 g/dose
Amoxicillin-clavulanate	AOM/sinusitis failure	45 mg/kg (amoxicillin component)	Oral	12-hourly	7–10 days	Diarrhoea
Ceftriaxone	Epiglottitis, mastoiditis, severe AOM failure	50–100 mg/kg	IV/IM	Once daily	7–14 days	2 g (4 g in meningitis)
Metronidazole	Mastoiditis, quinsy	7.5 mg/kg	IV/oral	8-hourly	10–14 days	500 mg/dose
Ciprofloxacin ear drops	CSOM	2–4 drops	Topical	3 times daily after wicking	2 weeks	Topical quinolones are safe in children
Dexamethasone	Croup	0.15–0.6 mg/kg	Oral/IM/IV	Once (repeat after 12–24 h if needed)	1–2 doses	12 mg (RCH); 16 mg in some guides
Prednisolone	Croup	1–2 mg/kg	Oral	Once	1 dose (repeat if needed)	–
Adrenaline 1:1000 (nebulised)	Severe croup	0.4–0.5 mL/kg	Nebulised	Repeat after 30 min if needed	–	**5 mL max**
Adrenaline 1:1000 (IM)	Anaphylaxis	0.01 mL/kg (0.01 mg/kg)	IM thigh	Every 5 min if needed	–	0.5 mg
Erythromycin	Penicillin allergy; diphtheria	12.5 mg/kg	Oral	6-hourly	10–14 days	500 mg/dose
Procaine benzylpenicillin	Diphtheria	50,000 IU/kg	IM	Once daily	Until can swallow (total 14 days)	1.2 MU
Diphtheria antitoxin	Diphtheria	20,000–100,000 IU by severity	IV over 1 h (or IM)	Once	–	Test dose; adrenaline ready
Rifampicin	Hib contacts	20 mg/kg	Oral	Once daily	4 days	600 mg

Complications

Infection	Complications
Common cold	AOM, sinusitis, pneumonia, asthma exacerbation, conjunctivitis
GAS tonsillitis	**Acute rheumatic fever → RHD**, post-streptococcal glomerulonephritis, quinsy, retropharyngeal abscess, scarlet fever, invasive GAS
AOM / CSOM	Perforation, **hearing loss**, speech delay, **mastoiditis**, **meningitis**, brain abscess, facial palsy, lateral sinus thrombosis, cholesteatoma
Sinusitis	Periorbital/orbital cellulitis, cavernous sinus thrombosis, frontal bone osteomyelitis (Pott's puffy tumour), brain abscess
Croup	Hypoxia, respiratory failure, bacterial tracheitis, pneumonia
Epiglottitis	Complete airway obstruction, death, Hib meningitis, pneumonia
Diphtheria	**Airway obstruction**, **myocarditis** (heart block, heart failure, sudden death), **neuropathy** (palatal palsy, diaphragm palsy), renal failure; case fatality about **30%** in unvaccinated people without treatment (WHO)

Ward-round & exam pearls

Model case presentation (child):

> "Sarah is a 6-year-old girl from Mukono weighing 19 kg (WAZ –0.6), fully immunised per UNEPI including pentavalent ×3, PCV ×3 and MR ×2, HIV-negative, on a family diet. She presents with 2 days of sore throat, painful swallowing and fever up to 39 °C, without cough or runny nose. No previous joint pains or known heart disease. On examination she is alert and drinking, T 38.8 °C, RR 22, HR 118, SpO_2 98%, with enlarged red tonsils with white exudate, tender anterior cervical nodes, uvula central, no membrane, no trismus; ears and chest normal; no murmur. McIsaac score 5. Assessment: **acute streptococcal tonsillitis**. Plan: benzathine penicillin 600,000 IU deep IM once (observed 30 minutes), paracetamol 280 mg (15 mg/kg) 6-hourly, fluids, counsel mother on the link with rheumatic fever and to return with joint pains, breathlessness or dark urine."

Likely viva questions with answers

1. *How do you distinguish croup from epiglottitis?* Croup: 6 m–3 y, prodrome, barking cough, hoarse voice, mild fever, not toxic, drinking. Epiglottitis: sudden high fever, toxic, drooling, tripod, muffled voice, little cough, won't swallow.
2. *What is the dose of dexamethasone for croup?* 0.15–0.6 mg/kg orally once (UCG: 0.15 mg/kg; severe: 0.6 mg/kg, max 12 mg).
3. *Why must a child given nebulised adrenaline be observed for hours?* Its effect lasts about 2 hours, after which stridor can return (rebound) while the steroid is still taking effect.
4. *Why treat strep throat for 10 days?* To eradicate GAS from the pharynx and **prevent rheumatic fever**; symptoms settle in 3 days but eradication needs 10 days (or one benzathine injection).
5. *Define CSOM and give its treatment.* Ear discharge ≥14 days through a perforation; dry wicking 3 times daily + topical quinolone drops for 2 weeks; no systemic antibiotics; test HIV, assess hearing.
6. *What is the first drug in suspected diphtheria?* Diphtheria antitoxin, as early as possible on clinical suspicion, plus penicillin/erythromycin, isolation and contact prophylaxis.

7. *What vaccine prevents epiglottitis, and when is it given in Uganda?* Hib, in the pentavalent DTP-HepB-Hib at 6, 10 and 14 weeks.

OSCE tips

- **Otoscopy station**: choose the largest speculum that fits; **pull the pinna down and back in infants, up and back in older children**; hold the otoscope like a pen with your little finger resting on the child's head; describe the drum (colour, light reflex, bulging/retraction, perforation).
- **Throat station**: "I would not examine the throat in a child with stridor or drooling". Say it before you pick up the spatula.
- **Counselling station (viral URTI)**: explain viral cause, expected duration (cough up to 2–3 weeks), home care, why no antibiotic, and the danger signs to return for.
- **Ear-wicking demonstration**: twist a clean cloth or tissue into a wick, insert gently, remove when wet, replace with a clean wick, repeat until dry, 3 times daily; never leave a wick in or use sticks/oils.

Mnemonics & memory aids

- **Croup "3 Bs and an S"**: **B**arking cough, **B**ad voice (hoarse), **B**etween 6 months and 3 years, **S**tridor.
- **Epiglottitis "4 Ds"**: **D**rooling, **D**ysphagia, **D**ysphonia (muffled voice), **D**istress (tripod), with toxicity.
- **FeverPAIN**: **F**ever, **P**urulence, **A**ttend rapidly, **I**nflamed tonsils, **N**o cough/coryza.
- **Centor "CENT"**: no **C**ough, **E**xudate, **N**odes (tender anterior), **T**emperature >38 °C (+ age for McIsaac).
- **Diphtheria "ABCDE"**: **A**ntitoxin, **B**ed rest (heart), **C**ulture and contacts, **D**rug (penicillin/erythromycin), **E**CG and **E**xclude airway obstruction.
- **Ear "<14 = acute, ≥14 = chronic"**.

High-yield summary

Numbers & doses to know

Item	Value
Colds per year (under 5)	**4–8**
Bacterial sinusitis	Symptoms **>10 days**, severe onset ≥3 days, or double sickening
Acute vs chronic ear discharge	**<14 days** vs **≥14 days**
GAS age range	**3–15 years** (rare <3 y)
Penicillin V	**10–20 mg/kg 6-hourly × 10 days**
Benzathine penicillin	**600,000 IU (<30 kg); 1.2 MU (≥30 kg)** IM once (UCG: 30,000 IU/kg if <30 kg)
Amoxicillin (AOM)	**40 mg/kg 12-hourly × 5 days** (WHO/IMCI); UCG 15 mg/kg 8-hourly
Topical quinolone (CSOM)	After wicking, 3 times daily × **2 weeks**
Mastoiditis	Ceftriaxone **50–80 mg/kg** daily + metronidazole **7.5 mg/kg** 8-hourly × 10–14 days
Dexamethasone (croup)	**0.15–0.6 mg/kg** once (max 12 mg)
Prednisolone (croup)	**1–2 mg/kg** once
Nebulised adrenaline 1:1000	**0.4–0.5 mL/kg, max 5 mL**; observe **2–4 h**
Ceftriaxone (epiglottitis)	**50–100 mg/kg** IV daily × 7–10 days
Hib contact prophylaxis	Rifampicin **20 mg/kg** daily × **4 days** (max 600 mg)

Item	Value
DAT	**20,000–100,000 IU** by severity (WHO pocket book 40,000)
Diphtheria antibiotics	**14 days**
Diphtheria myocarditis	Days **7–14**; neuropathy weeks **2–8**
UNEPI	DTP-HepB-Hib + PCV at **6, 10, 14 weeks**; MR at **9 and 18 months**

Classic exam traps

- **"Amoxicillin for a cold to prevent pneumonia."** No: no benefit, only harm.
- **"Look at the throat to confirm epiglottitis."** Never: it can cause complete obstruction.
- **"Five days of penicillin for strep throat."** In Uganda: **10 days** (or benzathine) to prevent rheumatic fever.
- **"Oral antibiotics for chronic ear discharge."** No: wick and use topical quinolone drops.
- **"Steam inhalation for croup."** No proven benefit, risk of scalds.
- **"Send home 30 minutes after adrenaline because the stridor has gone."** No: observe **2–4 hours** for rebound.
- **"Wait for the culture before giving diphtheria antitoxin."** No: give on clinical suspicion.
- **"Cough syrup for a 2-year-old."** UCG 2023: avoid cough mixtures under 6 years.

Questions seniors ask

This child has a red throat and a runny nose. Will you give an antibiotic? No. Cough and coryza point to a viral pharyngitis; I'd give paracetamol, fluids and advice on when to return.

What features make you think of streptococcal tonsillitis? Age 3–15, fever, tonsillar exudate, tender anterior cervical nodes, and no cough or coryza (McIsaac 4–5).

How do you treat moderate croup in an 18-month-old weighing 10 kg? Keep calm on mum's lap; dexamethasone 1.5–6 mg (0.15–0.6 mg/kg) orally once; observe; nebulised adrenaline 4–5 mL of 1:1000 if it becomes severe; oxygen if SpO_2 <92%.

What would make you think this "croup" is something else? Toxicity and high fever (epiglottitis, tracheitis), drooling (epiglottitis, retropharyngeal abscess), sudden onset without prodrome (foreign body), grey membrane (diphtheria), urticaria (anaphylaxis), or no response to adrenaline (tracheitis).

A 4-year-old has had ear discharge for a month. What do you do? CSOM: teach wicking 3 times daily, ciprofloxacin ear drops for 2 weeks, keep water out, HIV test, check hearing, ENT referral if not dry or if cholesteatoma/complications.

What is the role of diphtheria antitoxin and why the hurry? It neutralises circulating toxin before it binds to the heart and nerves; once bound it cannot be reversed, so give it on suspicion.

How does the pentavalent vaccine relate to this tutorial? It contains diphtheria toxoid and Hib conjugate, preventing diphtheria and Hib epiglottitis (and Hib meningitis/pneumonia); PCV prevents much pneumococcal AOM, sinusitis and pneumonia.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Management of the common cold in a 1-year-old (UCG 2023).	Continue breastfeeding, extra fluids, **saline nose drops** and gentle suction, paracetamol for 2–3 days, keep warm, return if danger signs. **No antibiotics, no cough syrups (<6 y), no honey (<1 y).**
2	IMCI classification 'No pneumonia: cough or cold' — what is it and what do you do?	Cough/cold **without** fast breathing, chest indrawing, stridor in a calm child or danger signs. Home care, soothe the throat, **no antibiotic**, return if worse, follow up in 5 days if not improving; cough >14 days → assess TB/asthma/pertussis.
3	McIsaac (modified Centor) score items.	Fever >38 °C (+1), tonsillar exudate/swelling (+1), tender anterior cervical nodes (+1), **absence of cough** (+1), age 3–14 y (+1), age ≥45 (–1). **4–5 = treat** for GAS.
4	FeverPAIN score.	**F**ever in past 24 h, **P**urulence, **A**ttend rapidly (≤3 days), **I**nflamed tonsils, **N**o cough or coryza. 0–1 no antibiotic; 4–5 antibiotic (NICE NG84).
5	First-line oral treatment of strep tonsillitis in a child (UCG 2023).	**Phenoxymethylpenicillin 10–20 mg/kg (max 500 mg) every 6 h for 10 days.** Penicillin allergy: erythromycin 12.5 mg/kg 6-hourly × 10 days (or azithromycin 12 mg/kg daily × 5 days).
6	Benzathine penicillin dose for strep tonsillitis.	**600,000 IU deep IM once if <30 kg; 1.2 MU if ≥30 kg** (WHO/AHA). UCG 2023: child <30 kg 30,000 IU/kg. Single dose; have adrenaline ready.
7	Why give a full 10-day course for GAS pharyngitis in Uganda?	To **eradicate GAS** and **prevent acute rheumatic fever/RHD** (treatment within 9 days of onset prevents most ARF). Uganda has a very high RHD burden. Post-streptococcal GN is not prevented.
8	Signs of a peritonsillar abscess (quinsy).	Severe unilateral throat pain, **trismus**, drooling, **muffled 'hot-potato' voice**, uvula pushed to the **opposite** side, fever, ear pain. Needs IV penicillin + metronidazole and **drainage** by ENT.
9	Define acute otitis media and give WHO/IMCI treatment.	Ear pain, fever, red **bulging** drum, or pus <14 days. **Amoxicillin 40 mg/kg twice daily × 5 days**, analgesia, dry by wicking, review in 5 days. (UCG 2023: 15 mg/kg 8-hourly × 5 days; AAP high-dose 80–90 mg/kg/day.)
10	Define CSOM and give its treatment.	Ear discharge **≥14 days** through a perforation. **Dry wicking 3×/day + topical quinolone (ciprofloxacin) drops for 2 weeks**; no systemic antibiotics; keep water out; HIV test; hearing check; ENT if persists.
11	How do you teach ear wicking?	Twist clean absorbent cloth/tissue into a wick, insert gently, remove when wet, replace with a clean wick, repeat until dry; **3 times daily**. Never cotton buds, sticks, oils or herbal drops.
12	Mastoiditis: signs and management (UCG 2023).	Tender swelling behind the ear, **pinna pushed forward and down**, fever, ear discharge. Admit; **ceftriaxone 50–80 mg/kg IV once daily + metronidazole 7.5 mg/kg 8-hourly × 10–14 days**; urgent ENT for drainage; watch for meningitis/abscess.
13	When do you diagnose bacterial sinusitis in a child?	Nasal discharge/daytime cough **>10 days** without improvement; **severe onset** (fever ≥39 °C + purulent discharge ≥3 days); or **double sickening**. Amoxicillin 7–10 days. Rare <5 y (UCG).

#	Question	Answer
14	Croup: typical age, cause and triad.	**6 months–3 years**; **parainfluenza** (also RSV, influenza, **measles**). **Barking cough, hoarse voice, inspiratory stridor**, after 1–2 days of coryza; worse at night.
15	Steroid doses for croup.	**Dexamethasone 0.15 mg/kg orally once** (UCG mild); **0.6 mg/kg (max 12 mg)** PO/IM/IV for severe (RCH). Or **prednisolone 1–2 mg/kg** once. Effect within 30 min–2 h.
16	Nebulised adrenaline for severe croup: dose and caution.	**Adrenaline 1:1000, 0.4–0.5 mL/kg (max 5 mL)** nebulised; repeat after 30 min if needed (UCG: 0.4 mg/kg max 5 mg). Effect lasts ~2 h: **observe at least 2–4 h** for rebound.
17	Westley croup score: components and cut-offs.	Consciousness (0/5), cyanosis (0/4/5), stridor (0–2), air entry (0–2), recession (0–3). **≤2 mild; 3–7 moderate; 8–11 severe; ≥12 impending respiratory failure.**
18	Epiglottitis: the classic picture.	Sudden **high fever**, toxic, **drooling**, **tripod/sniffing posture**, muffled voice, soft stridor, **little or no cough**, refuses to swallow. Classically **Hib** (rare since pentavalent vaccine).
19	Three things you must NOT do in suspected epiglottitis.	**Do not examine the throat** with a spatula, **do not lie the child down**, **do not give IM injections/take blood** before the airway is secured (UCG 2023). Keep with mother; airway in theatre.
20	Antibiotic for epiglottitis (UCG 2023).	**Ceftriaxone 50 mg/kg IV once daily for 7–10 days** (many centres 80–100 mg/kg, max 2 g), after the airway is secured. Rifampicin prophylaxis (20 mg/kg × 4 days) for household contacts if an unvaccinated child <4 y lives there.
21	Bacterial tracheitis: how does it present?	Starts like croup, then **sudden deterioration** with **high fever, toxicity, copious purulent secretions**, poor response to adrenaline. *S. aureus*/GAS. Airway (intubation) + IV antibiotics (e.g. ceftriaxone + cloxacillin).
22	Clinical features of diphtheria.	Unimmunised child, low fever, sore throat, **grey adherent pseudomembrane** that bleeds when removed, **bull neck**, hoarseness/stridor. Toxin → **myocarditis (days 7–14)** and **neuropathy** (palatal palsy, weeks 2–8).
23	Diphtheria treatment.	**Isolate**; **diphtheria antitoxin** on clinical suspicion (20,000–100,000 IU by severity; WHO pocket book 40,000); **procaine penicillin 50,000 IU/kg IM daily** then penicillin V (or erythromycin) to **14 days**; airway; daily ECG; notify; vaccinate after recovery.
24	Diphtheria contact management (UCG 2023).	Throat swabs; **benzathine penicillin once** (600,000 IU <10 y; 1.2 MU ≥10 y) or erythromycin 7–10 days; check/complete DTP, booster if last dose >1 year ago; monitor for 7 days.
25	Uganda UNEPI vaccines that prevent URTI complications.	**DTP-HepB-Hib (pentavalent) and PCV at 6, 10, 14 weeks** (diphtheria, Hib epiglottitis, pneumococcal AOM); **measles-rubella at 9 and 18 months** (measles croup/OM). HepB birth dose since 2025.
26	Unilateral foul-smelling nasal discharge in a toddler: diagnosis?	**Nasal foreign body** (bean, groundnut, foam, bead) until proven otherwise. Remove (refer to ENT if not easily removed); treat infection.
27	Why do toddlers get otitis media after a cold?	Their **Eustachian tube is shorter, wider and more horizontal**, and adenoids are large, so nasopharyngeal secretions and bacteria reach the middle ear. Smoke and bottle-feeding lying flat add risk; breastfeeding protects.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 3-year-old girl (14 kg) has had a runny nose, cough and mild fever for 3 days. She is playing, drinking well, RR 28/min, no chest indrawing, SpO_2 98%. The throat is mildly red with no exudate; ear drums are normal. Her mother asks for 'a strong antibiotic so it does not become pneumonia'. What is the MOST appropriate management?

- **A** Amoxicillin 40 mg/kg twice daily for 5 days to prevent pneumonia
- **B** Home care with fluids, saline nose drops and paracetamol, no antibiotic, and advice on danger signs
- **C** Cotrimoxazole for 5 days and a cough syrup
- **D** Benzathine penicillin 600,000 IU IM once
- **E** Admit for observation because she is under 5 years

2. A 10-year-old boy (28 kg) presents with 2 days of sore throat and fever 38.9 °C. He has no cough or runny nose. Both tonsils are enlarged with white exudate and there are tender anterior cervical nodes. There is no rapid strep test. His family lives 20 km away and his mother says he 'never finishes medicine'. What is the BEST treatment?

- **A** Paracetamol only, because most sore throats are viral
- **B** Amoxicillin for 3 days
- **C** Ciprofloxacin 15 mg/kg twice daily for 7 days
- **D** Benzathine penicillin 600,000 IU deep IM as a single dose
- **E** Prednisolone 1 mg/kg for 3 days

3. A 16-month-old boy (10 kg) wakes at 2 a.m. with a barking cough, hoarse cry and noisy breathing after 2 days of a runny nose. In the emergency room he is alert on his mother's lap, has inspiratory stridor at rest with mild subcostal recession, SpO_2 95%, T 37.9 °C. He is drinking. What is the MOST appropriate first treatment?

- **A** Oral dexamethasone 0.15–0.6 mg/kg once
- **B** Examine the throat with a spatula to exclude a foreign body
- **C** IV ceftriaxone 50 mg/kg
- **D** Salbutamol 2.5 mg nebulised every 20 minutes
- **E** Steam inhalation from a kettle for 20 minutes

4. Thirty minutes after oral dexamethasone, the 16-month-old with croup (10 kg) is agitated with loud stridor at rest, marked sternal and subcostal recession and SpO_2 90% in air. What is the MOST appropriate next step?

- **A** A second dose of oral dexamethasone and discharge
- **B** Immediate lateral neck X-ray
- **C** Diazepam 0.3 mg/kg rectally to calm him
- **D** IV ampicillin and gentamicin
- **E** Nebulised adrenaline 1:1000, 4–5 mL, with oxygen, and call senior/anaesthetic help

5. A 4-year-old unimmunised girl (13 kg) from a border district is brought in with high fever for 6 hours. She is sitting upright, leaning forward with her mouth open, drooling, with a soft inspiratory stridor and a muffled voice. She is not coughing. SpO_2 93%. What is the MOST appropriate immediate action?

A Lay her flat and examine the throat with a spatula and torch

B Insert an IV cannula and take blood cultures before anything else

C Keep her with her mother, give oxygen without distressing her, and call the most experienced anaesthetist/ENT to secure the airway in theatre

D Give IM ceftriaxone and send her for a lateral neck X-ray

E Give oral dexamethasone and observe in the waiting area

6. An 18-month-old boy (10 kg) has had fever and has been pulling at his right ear for 2 days after a cold. Otoscopy shows a red, bulging right tympanic membrane with no discharge. He is drinking and has no danger signs. Which treatment is MOST appropriate according to WHO/IMCI practice in Uganda?

A Ciprofloxacin ear drops for 2 weeks

B Wicking only, without antibiotics

C Ceftriaxone 50 mg/kg IV for 5 days

D Amoxicillin 400 mg (40 mg/kg) twice daily for 5 days with paracetamol

E Chlorphenamine and xylometazoline drops for 10 days

7. A 6-year-old HIV-positive boy (17 kg) on ART has had painless foul discharge from the left ear for 6 weeks. He has been given three courses of oral amoxicillin. There is no fever and no swelling behind the ear. Otoscopy shows pus and a central perforation. What is the BEST management?

A A fourth course of oral amoxicillin at a higher dose

B Dry mopping (wicking) three times daily plus ciprofloxacin ear drops for 2 weeks, keep water out, and arrange a hearing assessment

C Admit for IV ceftriaxone and metronidazole

D Insert cotton wool soaked in warm oil and review in a month

E Refer immediately for mastoidectomy

8. A 5-year-old girl (16 kg) with a 2-week history of right ear discharge now has fever 39.2 °C and a tender, fluctuant swelling behind the right ear that pushes the pinna forward and downward. She is alert with no neck stiffness. What is the MOST appropriate management?

A Admit, IV ceftriaxone 80 mg/kg daily plus metronidazole 7.5 mg/kg 8-hourly, and urgent ENT referral for drainage

B Oral amoxicillin and review in 5 days

C Wicking and ciprofloxacin ear drops only

D Incise the swelling in the OPD and send home on paracetamol

E Warm compresses and chlorphenamine

9. A 7-year-old unimmunised boy (20 kg) from a refugee settlement has had a sore throat and low-grade fever for 3 days. He has a markedly swollen neck and a grey-white membrane over both tonsils that bleeds when touched. His voice is hoarse. What is the MOST important immediate treatment?

A Oral amoxicillin and paracetamol

B Remove the membrane with forceps to relieve obstruction

C Nebulised adrenaline and discharge

D Wait for throat culture results before any specific treatment

E Isolate him and give diphtheria antitoxin plus penicillin, with airway monitoring

10. Ten days after admission for diphtheria, a 7-year-old who is now swallowing well suddenly becomes breathless with HR 46/min and a gallop rhythm. What is the MOST likely cause? Advanced

A Aspiration pneumonia
B Anaphylaxis to antitoxin
C Diphtheritic myocarditis with heart block
D Rheumatic carditis
E Hypokalaemia from penicillin

11. A 2-year-old with croup was given dexamethasone and nebulised adrenaline 40 minutes ago. Stridor at rest has disappeared, he is playing and SpO_2 is 97%. The mother wants to go home now. What should you advise?

A Go home now; the adrenaline has cured the croup
B Stay for observation for at least 2–4 hours after adrenaline; admit if stridor at rest returns
C Give a second adrenaline dose before leaving to prevent relapse
D Go home with oral amoxicillin to prevent bacterial tracheitis
E Stay for 24 hours of mandatory IV fluids

12. A 15-year-old girl (48 kg) has fever, severe sore throat with exudative tonsillitis, posterior cervical lymphadenopathy, marked fatigue and a palpable spleen tip. Which drug should you AVOID? Advanced

A Paracetamol
B Ibuprofen with food
C Salt-water gargles
D Penicillin V if GAS is confirmed
E Amoxicillin

13. A 9-year-old boy (25 kg) treated for tonsillitis with 3 days of amoxicillin now has severe left-sided throat pain, cannot open his mouth fully, is drooling, and speaks with a muffled voice. The left soft palate bulges and the uvula is deviated to the right. What is the MOST likely diagnosis and next step?

A Peritonsillar abscess: admit, IV penicillin plus metronidazole, and ENT drainage
B Epiglottitis: secure the airway in theatre before anything else
C Diphtheria: give antitoxin
D Viral pharyngitis: reassure and continue amoxicillin
E Dental abscess: extract the tooth in OPD

14. A 6-year-old girl has had a blocked nose with purulent discharge and daytime cough for 14 days without any improvement. She is afebrile with mild tenderness over the maxillary sinuses. There is no eye swelling. Which is the MOST appropriate treatment?

A No treatment; sinusitis is always viral
B Chlorphenamine and xylometazoline for 4 weeks
C IV ceftriaxone and urgent CT of the sinuses
D Amoxicillin for 7–10 days with saline irrigation and analgesia
E Ciprofloxacin orally for 14 days

15. A 3-year-old with croup has been in hospital for 2 days and was improving. Today he has a temperature of 39.8 °C, looks toxic, and has worsening stridor with thick secretions; two doses of nebulised adrenaline give little improvement. What is the MOST likely diagnosis? Advanced

A Rebound after adrenaline
B Epiglottitis
C Bacterial tracheitis
D Retropharyngeal abscess
E Laryngomalacia

Answers & explanations

1. B · Home care with fluids, saline nose drops and paracetamol, no antibiotic, and advice on danger signs

B is right: this is IMCI **'no pneumonia: cough or cold'**, a viral URTI; the treatment is home care, symptom relief and a clear safety net. A is wrong: antibiotics do not prevent pneumonia after a cold and cause harm. C is wrong: no antibiotic is indicated and UCG 2023 advises against cough syrups in children <6 years. D is wrong: there are no features of streptococcal tonsillitis (and GAS is uncommon at 3 years). E is wrong: she has no danger signs or distress; admission is unnecessary.

2. D · Benzathine penicillin 600,000 IU deep IM as a single dose

D is right: McIsaac score 5 (fever, exudate, tender nodes, no cough, age 3–14) → likely **GAS**. With doubtful adherence, a **single benzathine penicillin injection** (600,000 IU if <30 kg) guarantees eradication and prevents rheumatic fever. A is wrong: this is a high-probability bacterial tonsillitis in a high-RHD country. B is wrong: 3 days does not eradicate GAS; a 10-day course is needed. C is wrong: ciprofloxacin has poor GAS activity and is not indicated. E is wrong: steroids may relieve pain but do not treat infection or prevent rheumatic fever.

3. A · Oral dexamethasone 0.15–0.6 mg/kg once

A is right: this is **moderate croup** (stridor at rest, recession, alert). Every child with croup and stridor should get a single dose of **oral dexamethasone** (0.15 mg/kg UCG; up to 0.6 mg/kg). B is wrong: throat examination can upset the child and worsen obstruction, and the history (prodrome, barking cough) points to croup, not a foreign body. C is wrong: croup is viral; antibiotics are not indicated. D is wrong: salbutamol treats lower-airway bronchospasm, not subglottic oedema. E is wrong: humidified air has no proven benefit and steam kettles cause scalds.

4. E · Nebulised adrenaline 1:1000, 4–5 mL, with oxygen, and call senior/anaesthetic help

E is right: he now has **severe croup** with hypoxaemia. Give **nebulised adrenaline 1:1000 at 0.4–0.5 mL/kg (max 5 mL)** with oxygen, alert the airway team, and observe for at least 2–4 hours. A is wrong: steroids take time to work; discharging a hypoxic child is dangerous. B is wrong: X-rays delay treatment and move a child with a critical airway away from help. C is wrong: sedation can precipitate complete obstruction and respiratory arrest. D is wrong: there is no evidence of bacterial infection.

5. C · Keep her with her mother, give oxygen without distressing her, and call the most experienced anaesthetist/ENT to secure the airway in theatre

C is right: this is **epiglottitis** (toxic, drooling, tripod, muffled voice, no cough, unvaccinated). The priority is a calm child and a **secured airway** by the most skilled person, in theatre. A is wrong: lying flat and a spatula can cause complete obstruction. B is wrong: painful procedures before the airway is secure can precipitate obstruction (UCG: avoid procedures that agitate the child). D is wrong: IM injections should be avoided, and sending her to X-ray removes her from airway help. E is wrong: this is not croup; observation in a waiting area could be fatal.

6. D · Amoxicillin 400 mg (40 mg/kg) twice daily for 5 days with paracetamol

D is right: **acute otitis media** in a child <2 years; WHO pocket book/IMCI: **amoxicillin 40 mg/kg twice daily for 5 days** plus analgesia, review in 5 days. A is wrong: topical quinolones are for CSOM (discharge ≥14 days), and the drum is intact. B is wrong: wicking is for a discharging ear; this child needs an antibiotic and analgesia. C is wrong: IV ceftriaxone is for complications or treatment failure. E is wrong: antihistamines and decongestants do not treat AOM.

7. B · Dry mopping (wicking) three times daily plus ciprofloxacin ear drops for 2 weeks, keep water out, and arrange a hearing assessment

B is right: discharge ≥14 days through a perforation = **CSOM**: wicking + **topical quinolone** drops for 2 weeks; systemic antibiotics are not useful (UCG 2023); assess hearing and refer to ENT if it persists. A is wrong: repeated oral antibiotics fail in CSOM and breed resistance. C is wrong: there are no signs of complications (mastoiditis, meningitis) to justify IV therapy. D is wrong: oils and plugs keep the ear wet and infected. E is wrong: there is no mastoid swelling or tenderness; surgery is not the first step.

8. A · Admit, IV ceftriaxone 80 mg/kg daily plus metronidazole 7.5 mg/kg 8-hourly, and urgent ENT referral for drainage

A is right: this is **mastoiditis** with probable subperiosteal abscess: admit, **ceftriaxone 50–80 mg/kg + metronidazole** for 10–14 days (UCG 2023), and urgent ENT for drainage/mastoidectomy, watching for intracranial spread. B is wrong: oral treatment is inadequate for a bone infection with abscess. C is wrong: topical therapy cannot treat mastoid infection. D is wrong: drainage must be done by ENT under appropriate conditions with IV antibiotics; OPD incision risks facial nerve injury and incomplete drainage. E is wrong: these do nothing for a bacterial abscess.

9. E · Isolate him and give diphtheria antitoxin plus penicillin, with airway monitoring

E is right: an **unimmunised** child with an adherent grey membrane that bleeds and a **bull neck** has **diphtheria**. Antitoxin must be given **on clinical suspicion** (it neutralises only unbound toxin), with penicillin (or a macrolide) for 14 days, isolation, airway care, ECG monitoring and contact management. A is wrong: amoxicillin alone does not neutralise toxin. B is wrong: stripping the membrane causes bleeding and can worsen obstruction or toxin absorption. C is wrong: adrenaline does not treat diphtheria and discharge would be dangerous. D is wrong: waiting for culture loses the window for antitoxin.

10. C · Diphtheritic myocarditis with heart block

C is right: diphtheria toxin causes **myocarditis typically in days 7–14**, with heart block, arrhythmias and heart failure; daily ECG and strict bed rest are needed, with cardiac support. A is wrong: aspiration causes fever and focal chest signs, not bradycardia and gallop. B is wrong: anaphylaxis to antitoxin occurs within minutes to hours of the infusion, not 10 days later (serum sickness occurs at 7–10 days but causes rash, fever and arthralgia, not heart block). D is wrong: rheumatic fever follows GAS, not diphtheria. E is wrong: penicillin does not cause hypokalaemia of this kind.

11. B · Stay for observation for at least 2–4 hours after adrenaline; admit if stridor at rest returns

B is right: nebulised adrenaline works for about 2 hours and symptoms can **rebound**; observe at least 2–4 hours and admit if stridor at rest recurs. A is wrong: adrenaline is temporary; the steroid takes longer to act. C is wrong: adrenaline is given for current severe obstruction, not as prophylaxis. D is wrong: antibiotics don't prevent tracheitis and croup is viral. E is wrong: he is drinking; IV fluids are unnecessary.

12. E · Amoxicillin

E is right: this picture (teenager, exudative tonsillitis, **posterior** nodes, **splenomegaly**, fatigue) suggests **infectious mononucleosis (EBV)**; **amoxicillin** commonly causes a widespread maculopapular rash in EBV infection. A is wrong: paracetamol is safe symptomatic treatment. B is wrong: ibuprofen with food is reasonable for pain. C is wrong: gargles are harmless symptomatic care. D is wrong: penicillin V can be used if co-existing GAS is proven; it is far less likely than amoxicillin to cause the EBV rash. (She should also avoid contact sports while the spleen is enlarged.)

13. A · Peritonsillar abscess: admit, IV penicillin plus metronidazole, and ENT drainage

A is right: unilateral pain, **trismus**, drooling, muffled voice and a **bulging palate with uvula pushed to the opposite side** = **quinsy**; needs IV antibiotics (penicillin + metronidazole) and **needle aspiration or incision** by ENT. B is wrong: epiglottitis is a midline supraglottic infection without trismus or palatal bulge. C is wrong: there is no membrane and he is presumably immunised. D is wrong: these are not features of a simple viral sore throat. E is wrong: the findings are in the tonsillar fossa, not a tooth.

14. D · Amoxicillin for 7–10 days with saline irrigation and analgesia

D is right: symptoms **>10 days without improvement** meet criteria for **acute bacterial sinusitis**: amoxicillin for 7–10 days (UCG 2023), saline irrigation and analgesia. A is wrong: persistent symptoms beyond 10 days suggest a bacterial infection that benefits from antibiotics. B is wrong: decongestant drops should not be used beyond 5 days (rebound) and antihistamines don't treat infection. C is wrong: IV therapy and CT are for orbital or intracranial complications. E is wrong: ciprofloxacin is not first line for pneumococcus/H. influenzae sinusitis in children.

15. C · Bacterial tracheitis

C is right: croup that **suddenly worsens** with **high fever, toxicity, purulent secretions** and **poor response to adrenaline** suggests **bacterial tracheitis** (S. aureus, GAS): secure the airway and give IV antibiotics. A is wrong: rebound occurs within hours of adrenaline and does not cause high fever or toxicity. B is wrong: epiglottitis begins abruptly, not after days of croup, and children have little cough. D is wrong: retropharyngeal abscess causes neck stiffness, drooling and torticollis, typically without a croup prodrome. E is wrong: laryngomalacia is chronic stridor from early infancy, not an acute febrile illness.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Night-time stridor at Mengo emergency

Daniel, a 14-month-old boy, is brought at 1 a.m. with a 'cough like a dog' and noisy breathing that began tonight after 2 days of a runny nose and mild fever. He is on his mother's lap, crying intermittently. His immunisation card shows BCG, OPV, pentavalent ×3, PCV ×3 and MR1 at 9 months.

1. What is your immediate assessment, and what will you avoid?

> ETAT: look for **stridor at rest, recession, cyanosis, SpO_2, consciousness, drooling**. Keep him **on his mother's lap**, examine gently, **avoid** throat examination with a spatula, lying him flat, painful procedures and separation from the mother.

2. Findings: weight 9 kg (WAZ −0.9), T 38.0 °C, RR 40, HR 145, SpO_2 94%, inspiratory stridor at rest with moderate subcostal recession, alert, no drooling, drinking. Classify.

> **Moderate croup** (stridor at rest, recession, alert; Westley about 4). Probable viral (parainfluenza). Not epiglottitis (no toxicity, drooling or tripod; he has a barking cough and is drinking).

3. Write the initial treatment.

> **Dexamethasone 0.15–0.6 mg/kg orally once** (1.4–5.4 mg; UCG 0.15 mg/kg; many units give 0.6 mg/kg) or prednisolone 1–2 mg/kg; paracetamol 135 mg (15 mg/kg); keep calm; observe SpO_2 and work of breathing every 15–30 min. No antibiotics, no steam, no cough syrup.

4. Forty minutes later he is agitated, stridor is louder, recession marked, SpO_2 89%. What now?

> **Severe croup**: **nebulised adrenaline 1:1000, 0.5 mL/kg = 4.5 mL** (max 5 mL), oxygen held near the face, call the senior and anaesthetist; repeat adrenaline after 30 min if needed; prepare for intubation with a tube 0.5–1 mm smaller than usual if he tires.

5. He improves after one adrenaline nebulisation (no stridor at rest, SpO_2 96%). Disposition?

> **Admit** (needed adrenaline, age 14 months, hypoxaemia at presentation, night time). Observe for **rebound** for at least 2–4 h; continue oral fluids; repeat steroid at 12–24 h if stridor persists; oxygen if SpO_2 <92%.

6. On day 2 the mother asks whether he needs 'the measles injection' and antibiotics. How do you respond?

> Croup is viral; **antibiotics are not needed** unless signs of bacterial tracheitis or pneumonia appear. Check for measles features (rash, conjunctivitis, Koplik spots); his **MR2 is due at 18 months** (UNEPI). Discharge when no stridor at rest and drinking, with advice to return for stridor at rest, recession, drooling or blue colour.

Take-away: Croup is graded by stridor at rest, recession, SpO_2 and consciousness: every child gets a steroid, severe croup gets nebulised adrenaline with oxygen, and anyone given adrenaline is observed for rebound.

Case 2: A sore throat and a heart at risk

Grace, an 11-year-old girl from Mbale, comes to the HC IV with 2 days of sore throat, fever and painful swallowing. Her older brother has rheumatic heart disease and receives monthly injections at the regional referral hospital.

1. What features in the history and examination would point to streptococcal tonsillitis rather than a viral sore throat?

> Age 3–15, **fever >38 °C**, **tonsillar exudate**, **tender anterior cervical nodes**, **no cough or coryza**, palatal petechiae, scarlet rash. Viral clues: cough, runny nose, hoarseness, conjunctivitis, ulcers, diarrhoea.

2. She is 32 kg, T 38.7 °C, exudative tonsils, tender anterior nodes, no cough, no murmur. Score her and decide.

> **McIsaac 5** → high probability of GAS. In Uganda (high RHD burden, family history of RHD, no rapid test): **treat**.

3. Prescribe treatment, with dose and duration.

> Either **benzathine penicillin 1.2 MU deep IM once** (≥30 kg), or **penicillin V 500 mg (max dose) 6-hourly for 10 days** (UCG 10–20 mg/kg/dose). Penicillin allergy: erythromycin 12.5 mg/kg (400 mg) 6-hourly × 10 days. Plus paracetamol 15 mg/kg (480 mg) 6-hourly and fluids. Observe 30 min after the injection with adrenaline ready.

4. Why does the duration matter, and what do you tell the family?

> A full **10-day course (or the injection)** eradicates GAS and **prevents acute rheumatic fever**; stopping when the pain settles (day 2–3) leaves bacteria behind. Tell them to **return if joint pains, breathlessness, chest pain, jerky movements or dark urine/puffy face** appear in the next weeks.

5. Three days later she cannot open her mouth fully, is drooling, and her voice is muffled; the right palate bulges. What has happened and what do you do?

> **Right peritonsillar abscess (quinsy)**. Admit, IV fluids, **IV benzylpenicillin 50,000 IU/kg 6-hourly + metronidazole 7.5 mg/kg 8-hourly**, analgesia, and **refer to ENT for aspiration/incision and drainage**; complete 10–14 days of antibiotics.

> **Take-away:** In Uganda, a high-scoring sore throat in a school-age child is treated with a full 10 days of penicillin or a single benzathine injection, because what you are really preventing is rheumatic heart disease.

Case 3: The drooling child from the border

Amina, a 5-year-old girl, arrives from a refugee settlement in Isingiro with high fever since this morning. She is sitting on her father's lap, leaning forward with her chin out and mouth open, drooling. She has never been vaccinated.

1. What are your top two diagnoses and what single action protects her most?

> **Epiglottitis** (unvaccinated, toxic, drooling, tripod) and **diphtheria** (unvaccinated; membrane, bull neck); also retropharyngeal abscess and bacterial tracheitis. The single most protective action: **keep her calm and upright with her father, oxygen without distress, and call the most experienced anaesthetist/ENT now**.

2. What must you not do?

> Do **not** lie her down, examine the throat with a spatula, insert cannulae or give IM injections, or send her alone for an X-ray before the airway is secured (UCG 2023).

3. In theatre the anaesthetist sees a cherry-red, swollen epiglottis and intubates her. What treatment follows?

> IV access, blood culture, **ceftriaxone 50–100 mg/kg IV once daily for 7–10 days** (16 kg → 800 mg–1.6 g; UCG 50 mg/kg), ICU/HDU care until the swelling settles (usually 24–48 h), sedation while intubated, then extubation in a controlled setting.

4. Her 2-year-old brother, also unvaccinated, lives in the same tent. What do you do for the family?

> **Rifampicin prophylaxis 20 mg/kg (max 600 mg) once daily for 4 days** for household contacts (an unvaccinated child <4 y is present); **start catch-up vaccination** (pentavalent with Hib, PCV, MR, OPV/IPV) for both children; notify the settlement health team.

5. How would the picture differ if this were diphtheria, and what would you add?

> Diphtheria: **slower onset**, low-grade fever, **grey adherent membrane** that bleeds when removed, **bull neck**, hoarseness. Add **isolation**, **diphtheria antitoxin** (20,000–100,000 IU by severity) on suspicion, penicillin or erythromycin for 14 days, **daily ECG** (myocarditis days 7–14), contact prophylaxis, notification, and vaccination.

Take-away: A drooling, toxic, unvaccinated child needs a calm parent, oxygen and the best airway person, not a spatula; then antibiotics, contact prophylaxis and the vaccines she missed.

References

1. Uganda Clinical Guidelines 2023 (5.2.3 Coryza; 5.2.4 Epiglottitis; 5.2.7 Croup; 21.1.4 Otitis media; 21.1.6 Mastoiditis; 21.2.4 Sinusitis; 21.3.3–21.3.5 Pharyngitis, tonsillitis, quinsy; 2.1.3 Diphtheria). *Ministry of Health Uganda*, 2023. https://library.health.go.ug/uganda-clinical-guidelines-2023
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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.